Mirpur University of Science and Technology

Department of Software Engineering

Project Report

Name: Haseeb Ahmad, Ali Usman

Roll No: 027, 007

Section: A

Session: 23-27

Subject: Web Engineering

Submitted to: Engr. Shamila Nasreen

Project Details

Project Title: SkillSwap – Peer-to-Peer Mentorship Platform

SkillSwap is a full-stack web application designed to facilitate peer-to-peer mentorship among students. It allows users to register their profiles, list skills they are willing to share (such as photography or programming), and view a community dashboard of available mentors. The primary objective is to create a collaborative learning environment through technology.

System Architecture & Folder Structure

The application follows a standard **Client-Server-Database** architecture. The **Frontend** (HTML/CSS/JS) communicates with a **Node.js/Express** backend via RESTful API calls. The backend handles the business logic and performs CRUD operations on a **MySQL** relational database.

Name	Date modified	Type	Size
backend	03/03/2026 5:55 pm	File folder	
database	03/03/2026 5:55 pm	File folder	
frontend	03/03/2026 7:12 pm	File folder	
node_modules	03/03/2026 7:03 pm	File folder	
.gitignore	03/03/2026 9:42 pm	Git Ignore Source ...	1 KB
package.json	03/03/2026 7:03 pm	JSON Source File	1 KB
package-lock.json	03/03/2026 7:03 pm	JSON Source File	34 KB
README.md	03/03/2026 9:20 pm	Markdown Source...	3 KB

Database Schema and Relational Design

The project utilizes a relational database to ensure data integrity. It consists of two primary tables:

- **Users Table:** Stores basic profile information (Name, Email).
- **Skills Table:** Stores the skills offered, linked to the Users table via a **Foreign Key** (user_id).

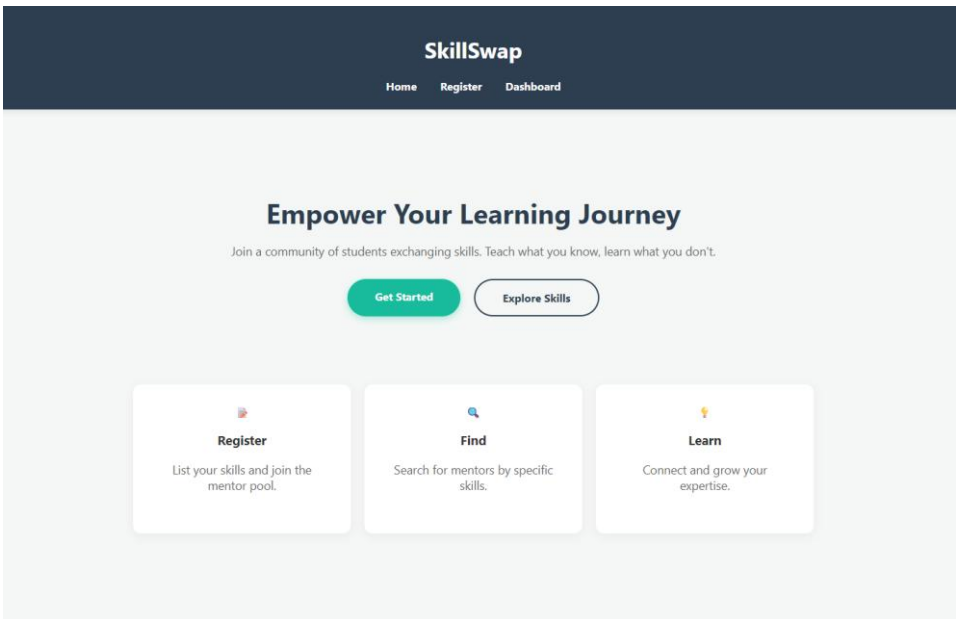
This relational structure allows for **Cascade Deletes**, meaning if a user is removed, their associated skills are automatically cleaned up.

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
name	varchar(100)	NO		NULL	
email	varchar(100)	NO	UNI	NULL	

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
user_id	int	YES	MUL	NULL	
skill_name	varchar(100)	NO		NULL	

Application Interface & Navigation

The UI is designed to be clean and professional. The landing page features a high-impact **Hero Section** with clear "Call to Action" buttons to guide the user. The registration form is centered for better focus and follows modern design principles.

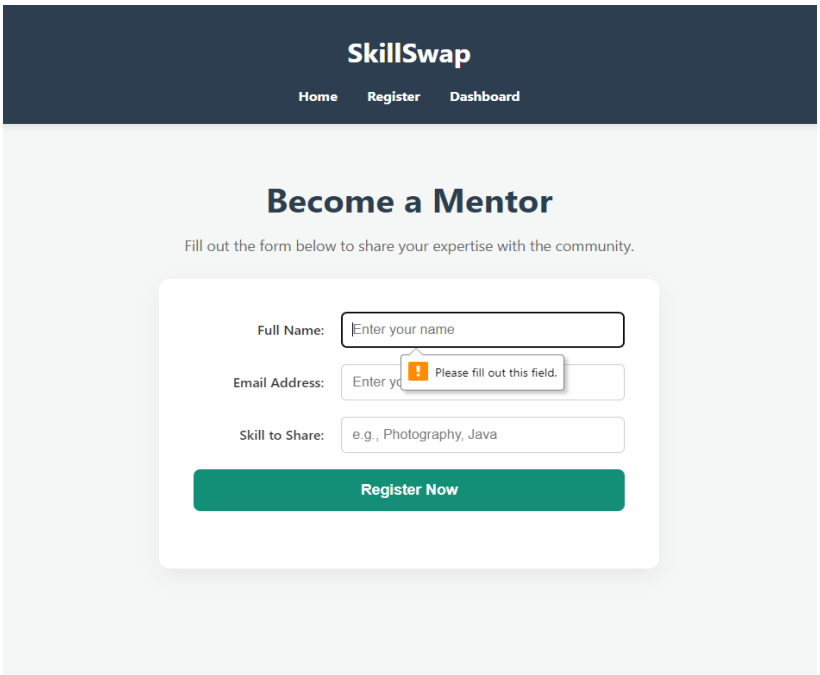


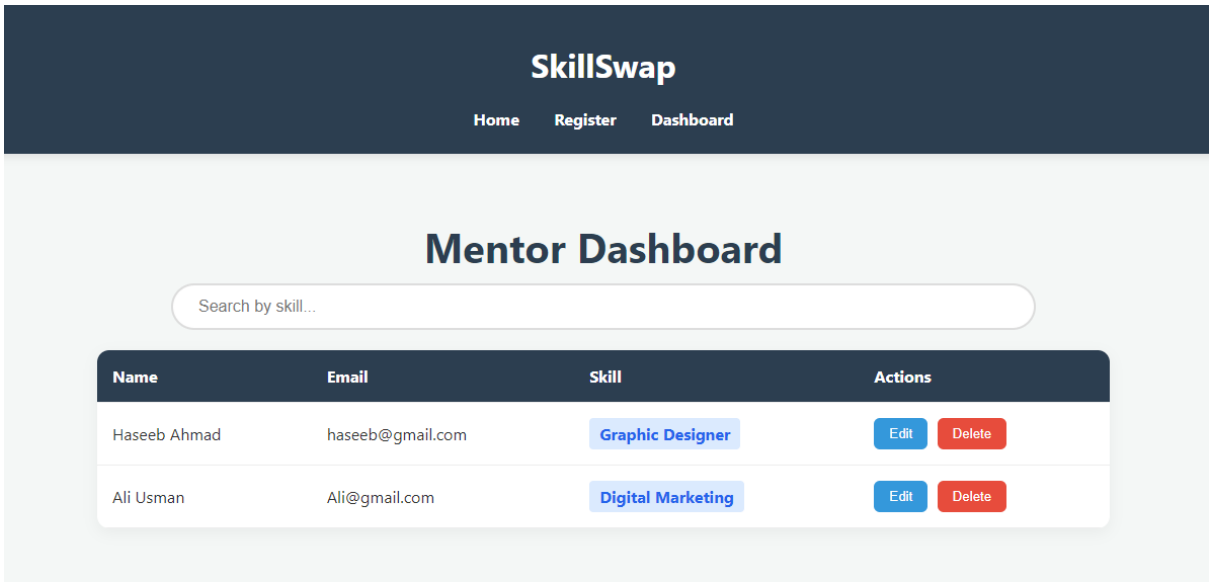
CRUD Functionality and Client-Side Validation

The system implements full CRUD functionality:

1. **Create:** Users register via a validated form.
2. **Read:** The dashboard dynamically fetches and displays mentor data.
3. **Update:** Users can modify their skills via a secure prompt.
4. **Delete:** Profiles can be removed with a confirmation safety check.

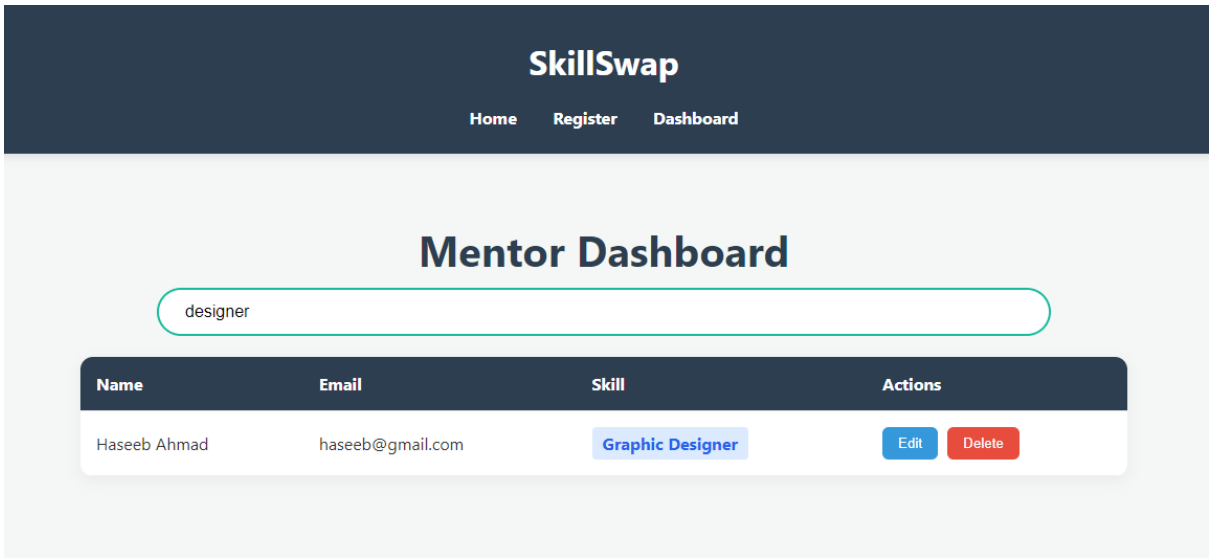
Client-Side Validation: JavaScript ensures that no empty fields are submitted to the server, providing immediate feedback to the user.





Real-Time Search Functionality

To enhance user experience, a real-time search filter is implemented on the dashboard. This allows students to quickly find mentors by typing a specific skill (e.g., "Photography") without reloading the page.



Conclusion and Future Enhancements

Developing **SkillSwap** provided practical experience in full-stack integration, specifically connecting a Node.js environment to a relational MySQL database. Future enhancements could include user authentication (Login/Password), a rating system for mentors, and a direct messaging feature for students to coordinate their swap sessions.

GitHub Repo Link

https://github.com/ahmad2297/SkillSwap.git